

Estimating Varietal Proportions of Wheat Grain Varieties with Fine-Tuned Deep Vision Models

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Abstract

Agroecology is becoming increasingly necessary in the face of climate change and biodiversity collapse. Multivarietal cropping is a promising approach, but understanding its mechanisms requires estimating the yield of each variety, i.e., classifying grains after harvest. In this paper, we propose (1) a real-world RGB dataset of 26,882 wheat grain images from 8 varieties, collected from our field trial in Saclay, France, and (2) a practical approach using recent deep learning models. We achieve high accuracy (up to 94%) in grain variety classification, and we show that fine-tuning the entire network outperforms using pretrained models as fixed feature extractors, even when source and target domains are very similar. We further evaluate our method by simulating mixed cropping scenarios and estimating varietal proportions at harvest time. The resulting performance indicates that the approach is sufficiently accurate for practical use.

Keywords: Agroecology, Supervised learning, Neural networks and deep learning, Transfer learning..

1 Introduction

1.1 Motivation, Context

Agriculture is facing the combined threat of climate change and biodiversity collapse (Hultgren et al., 2025; Leroy et al., 2026; Dannenberg et al., 2024). It therefore needs to adapt to an increasingly variable environment, notably by diversifying its cultivated communities to benefit from a portfolio effect (Loreau et al., 2021; Paut et al., 2019). Agricultural practices also need to change to attenuate their own impact on ecosystems (Moss, 2008; Sutton et al., 2011). To achieve both adaptation and attenuation, diversification at all spatio-temporal scales is a key lever of the agroecological transition (Mijatović et al., 2013; Ponisio et al., 2015; Beillouin et al., 2021).

Among diversification practices, cultivar mixtures consist in sowing in the same plot multiple varieties of the same species. They not only provide an increase in yield and stability (Huang et al., 2024), but also avoid major lock-in situations by being easy to implement for farmers without disturbing too much their supply chain. Their benefits in controlling epidemics of airborne diseases are

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well-documented (Mathieu et al., 2025), but their performance, although positive on average, display a large variance (Borg et al., 2018).

Understanding the mechanisms underlying the variance in mixture performance requires comparing the yields of numerous varieties grown in monovarietal conditions and in various mixtures across diverse environments. However, at the time of harvest, only the *total* mixture yield is available. The proportion of grains harvested per variety *within* each mixture remains unknown, a proportion that is not necessarily the same as at the time of sowing. Yet, such changes in proportions between sowing and harvest would inform us about the ability of a particular variety to dominate its neighbors or, conversely, to promote complementarity. This would be interesting not only from a fundamental perspective in terms of the evolution of populations in fluctuating environments, but also from an applied perspective for a range of agricultural and agri-food stakeholders (breeders, cooperatives, sorters, millers, etc).

Different varieties being of the same species, their grains are hard to distinguish. Field trials for research purposes relied on manual sorting (Finckh and Mundt, 1992; Stefan et al., 2025), but this is obviously not transposable to a farmer’s context. The ultimate alternative would consist in directly sequencing the DNA extracted from a sample of grains. However, such a destructive procedure still requires a grinding step that is hard to parallelize. We hence set up a dedicated experiment to explore the feasibility of discriminating grains based on their spectral properties, or here simply based on RGB pictures.

In this context, using recent Deep Learning techniques appears to be a promising approach. In this work, we build a dataset of wheat grain images (1 grain per image), including 8 wheat grain varieties, and perform multi-class classification. Although we have access to the hyperspectral (HSI) data, we restrict ourselves to the RGB bands. One may expect yet higher accuracies when using the full data, however, there are several advantages to relying on standard RGB images. First, the cost of data acquisition can be greatly reduced, as HSI cameras are very expensive. Second, the data management becomes easier (reducing from 216 to 3 channels represents a 72 fold reduction in dataset size). Third, we show in this paper that for all practical purposes, the RGB data is enough to obtain good accuracy. Lastly, this allows us to use highly optimized Computer Vision architectures and training pipelines, and to initialize networks with pre-trained weights coming from training on a large dataset (ImageNet (Deng et al., 2009), 10^6 images).

In this work, we therefore propose an RGB-based approach for wheat variety classification and evaluate its effectiveness.

1.2 Contributions

Our main contributions are:

- RGB Wheat Grain Dataset: We introduce a real-world dataset of 26 882 RGB images ($224 \times 224 \times 3$) of individual wheat grains from 8 varieties, collected over two growing seasons under controlled field conditions. The dataset is released as a public benchmark ¹ to encourage reproducibility and community contributions.
- Empirical proof that to achieve very high (94%) and robust accuracies: (i) RGB inputs are sufficient; (ii) about 1600 samples per class are enough; (iii) the training strategy, in particular, data augmentations, is important.
- Introduction of a task-tailored calibration protocol that shows our best model estimates harvest-time varietal proportion very well and provides uncertainty estimates.
- Transfer Learning: lightweight fine-tuning of our model’s head is shown to display high performance (89% accuracy), demonstrating the robustness of the learned representation and allowing

¹https://www.codabench.org/competitions/15153/?secret_key=696aea33-8087-4cd7-9443-c3290b541235

easy extension to new grain data even under limited compute/label resources. This also confirms the superiority of fine-tuning the entire network (94% accuracy).

2 Dataset

2.1 Data acquisition

A field trial was conducted on the “plateau de Saclay”, France, in the frame of the PerfoMix project. Eight varieties of bread wheat were cultivated in monovarietal conditions (often called “pure” stand) and in mixtures composed of two to four varieties (mixed stand). More details about the eight varieties can be found in (Gawinowski et al., 2024). The field trial was carried out in 2019-2020 (Year 1, Y1) and was repeated in 2020-2021 (Year 2, Y2). The two seasons were conducted on the same soil type (different plots of the same farm), under the same agricultural practices (comprising two nitrogen fertilization events and two fungicide applications per season), so the main difference came from the climatic conditions of those years. Each year, each variety was cultivated on two micro-plots of approximately 7 square meters. Each plot was harvested (grains collected from plants) separately from other plots, and grain bags were labeled according to which variety (or varieties) was sown in that plot.

A sample of grains from each plot was analyzed using hyperspectral imaging (HSI). This technique provides a 2D image with, for each pixel, the reflectance level at several hundred wavelengths in the visible and near-infrared, known as a “spectrum”. Concretely, instead of the three RGB channels of a classical image, the HSI images have more than 200 channels. In practice in this work we discarded the HSI aspect of data and retained only the bands corresponding to the Red, Green, and Blue channels (RGB).

Thanks to the monovarietal plots (pure stand), we obtained labeled data relatively easily: by construction, all grains collected from a given plot were from the variety that was sown in that plot. It is worth mentioning that, for scientific purposes, such control plots (monovarietal) are very often present in field trials, so that labeled data are usually available for each variety and set of conditions (soil and climate).

2.2 Data pre-processing

For practical reasons, for each variety and year, hundreds of grains were scattered across a black (clean) background and photographed at once. Each such raw acquisition covered 2048×9100 pixels across 216 spectral bands (visible to near-infrared). A watershed segmentation algorithm (Price-Whelan, 2017; Price-Whelan et al., 2025) was then used to detect grains from the background and output individual-grain smaller images (shape $252 \times 252 \times 216$). It can happen that some grains are so close together that they are detected as one grain, however this is relatively rare. For training data it is not a problem, as training occurs by taking crops of the input image anyway. For test data, it is important to scatter grains sufficiently so that this very rarely happens, otherwise the model will not be able to predict the varieties reliably, in the case where the 2 grains close together are from 2 distinct varieties.

Overall, we extract 26882 individual grains from these scene-level hyperspectral images, corresponding to about 1600 wheat grains for each of the $K = 8$ varieties, for each year (2 years). See appendix Fig. 2 to see how the dataset breaks down by variety and year.

From the HSI input (216 channels), three bands are selected for their ability to match usual RGB input: band 22 (≈ 680 nm, visible red / chlorophyll absorption), band 53 (≈ 550 nm, green), and band 89 (≈ 800 nm, blue). After this pre-processing, the dataset ends up being relatively small: $(26882 \times 256 \times 256 \times 3)$ integer values only take about 1.2 Go to store.

Some operations are done on-the-fly at data loading. Raw int16 values are divided by per-channel scale factors $\mathbf{s} = [1567, 8316, 18126]$ (fit on training percentiles) and clipped to $[0, 1]$, then standardized

to ImageNet mean and standard deviation. The same normalization is applied to both years and all architectures. The networks also resize the input samples into $224 \times 224 \times 3$ (float32) inputs (using bilinear interpolation).

3 Methodology

3.1 Architectures

In this work, we explored five architectures in order to find the best under/overfitting compromise. Here we quickly summarize the distinctive traits of each of these five architectures and briefly allude to why others were excluded.

ResNet (He et al., 2016) introduced residual connections to avoid the problem of the vanishing gradient and became the foundation of deep learning vision models. It remains a robust computer vision baseline and the dominant baseline in applied agricultural imaging (Guo et al., 2024). EfficientNetV2 (Tan and Le, 2021) introduced two novelties: 1) Fused-MBConv, which speeds up training by merging multiple convolution steps into one, reducing overhead and making better use of hardware like GPUs; 2) compound scaling improves efficiency by increasing depth, width, and input resolution in a balanced way, avoiding wasted computation. Vision Transformer (ViT) (Dosovitskiy et al., 2021) was inspired by the success of transformers in text processing and applies the idea of self-attention over tokens, where a token corresponds to a patch of 16×16 pixels. Swin Transformer (Liu et al., 2021) resolved the problem of ViT’s attention layer having quadratic complexity through hierarchical shifted windows, a method that was used in agriculture for pest control (Li et al., 2025). ConvNeXt (Liu et al., 2022) demonstrated that purely convolutional networks equipped with Transformer-inspired design choices (large-kernel depthwise convolutions, LayerNorm, GELU) can match Transformer accuracy while retaining convolutional inductive biases. Admittedly, ConvNeXt performs operations equivalent to those of a transformer but without explicitly using transformer layers. We summarize the ConvNeXt-Tiny model in appendix Fig. 4.

Other architectures such as DenseNet (Huang et al., 2017) and MobileNetV3 (Howard et al., 2019) were not considered due to their respective limitations in this context: DenseNet’s high memory consumption and computational overhead, and MobileNetV3’s reduced capacity for capturing fine-grained visual features.

For all the models we consider, we use ImageNet-pretrained weights as starting point, then fine-tune the entire network (backbone included) on our data. We mostly use the smallest variants available, see Table 1 for the model’s full names and our code for the exact implementation.

3.2 Training Methodology

Our training protocol (fine-tuning, really) is inspired by well-established, relatively recent methods in convolutional neural network training methods (He et al., 2019; Wightman et al., 2021; Lee et al., 2020; Vryniotis, 2021). Specifically, we adopt the modern baseline established by (Wightman et al., 2021), which advocates for the AdamW optimizer combined with a cosine annealing schedule. Following (He et al., 2019) and (Lee et al., 2020), we compound multiple data augmentation and regularization techniques to significantly boost the effective dataset size. Moreover, we have also integrated advanced primitives such as Stochastic Weight Averaging (SWA) as recommended (Vryniotis, 2021).

We report the full details of our training protocol in appendix A.2. The architecture (ConvNeXt) and training pipeline are summarized in appendix Fig. 4. We display a typical training run, namely the evolution of Bal-Acc (balanced accuracy) over epochs, in Appendix Fig. 5-right. We also report results obtained without the data augmentations, in Appendix Fig. 5-left, showing that those are indeed important for optimal generalization. Further ablations of some of our training details show that training is rather robust to other changes, see appendix Table 3.

Model	Y1 Native $\uparrow$	Y2 Native $\uparrow$	Compute $\downarrow$	Param. Count $\downarrow$
ConvNeXt-Tiny	0.944 ± 0.003	0.945 ± 0.002	≈ 5 min	28.6 M
EfficientNetV2-S	0.934 ± 0.003	0.938 ± 0.002	≈ 5 min	24.0 M
Swin-T	0.935 ± 0.002	0.931 ± 0.003	≈ 4 min	28.3 M
ViT-B/16	0.908 ± 0.003	0.911 ± 0.002	≈ 6 min	86.6 M
ResNet-18	0.853 ± 0.008	0.860 ± 0.004	≈ 5 min	11.2 M

Table 1: **Comparison of models: Bal-Acc.** Models are sorted by Y1 Native balanced accuracy. Results are reported using 5-fold Cross-Validation, with standard deviation as an uncertainty estimate. The compute time refers to the training (fine-tuning) time on an NVIDIA RTX PRO 6000 Blackwell (Server Edition).

4 Results

The first task, which is our main interest, is to classify grain varieties from a given year after training on data from the same year. Independence of train and test set comes from the fact that each grain is exclusively in one of the two splits. This is what we call the Native task: Native Y1 and Native Y2.

4.1 Robustness

Our metric is balanced accuracy (Bal-Acc), to avoid artifacts from the slight class imbalance. In appendix B.4 we report the full confusion matrix (applied to the test set) for our best model, ConvNeXt. No particular confusion trend appears in the normal tasks, thus validating Bal-Acc as a good choice of summary statistics.

We are in a relatively small data regime: with $K = 8$ classes and 1600 images per class compared to millions of parameters, we are in the overparameterized regime of Deep Learning. Thus, we control for lucky results by performing cross validation: for 5 different train/test splits, we train the model (splitting again train in train+validation each time), and measure the 5 test scores. We report all test Bal-Acc in appendix, Table 2 and summarize them in Table 1. Finding standard deviations that are mostly of the order $\approx O(0.003)$, we conclude that we are robust against the training set sampling noise, despite being in a relatively low data regime. In addition, this careful check shows that ConvNeXt performs significantly better than all the other models.

4.2 Comparison of Architectures

Table 1 summarizes our comparison of architectures. ConvNeXt-Tiny and EfficientNetV2-S are standing out, and ConvNeXt is statistically significantly better. They are closely followed by Swin-T. For ConvNeXt, we consistently obtain a Bal-Acc of 94%. Since ConvNeXt-Tiny is our best performing model, in the remaining of the paper, we focus our analysis on that model’s predictions.

4.3 Calibration

For ConvNeXt, we report the calibration of the model, *i.e.* its confidence (as measured by the largest logit value, among $K = 8$) is compared with its actual probability of being correct. Note that since we apply TTA, a lot of noise has already been removed. Concretely, for 8 bins of model’s self-confidence (from 0.2 to 1.0, by steps of 0.1), we measure the fraction of correct predictions, and the alignment is nearly perfect (plot is reported in Appendix, Fig. 7). There are ways to re-calibrate ML models, here calibration is spontaneously very good, so we do not need to apply any corrective.

For completeness, the distribution of confidence levels is also reported in appendix, Fig 8. We notice it is concentrated in the range $[0.9, 0.97]$, where the model predicts perfectly.

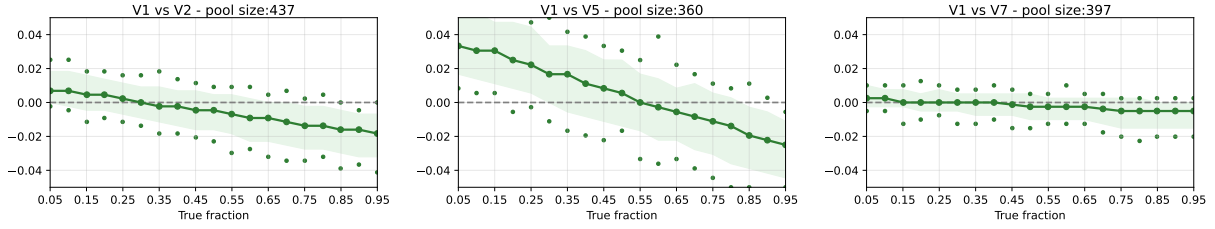


Figure 1: **Varietal proportion Estimates** A positive error indicates the first variety of the pair has been over-estimated. Bold line is the median over bootstrapped test sets. Shaded area indicates the 95% confidence interval. Min and Max value are shown as outlying smaller dots. Some pairs are perfectly estimated. The worse error is of about $\pm 4\%$ at the tips of the distribution, which is a large relative error in those cases, but the absolute error remains low.

4.4 Varietal proportion estimates

To test our model in the real application case’s conditions, we simulate possible harvest-time varietal proportions of all varieties pairs and study the absolute error.

Concretely, for all possible variety pairs (i, j) and for all varietal proportions from $p_i, p_j = 0.05, 0.95$ to $p_i, p_j = 0.95, 0.05$, we consider a number of test samples $N_{pool} = \min(N_i, N_j)$ and construct a simulated test set by sampling (with replacement) $p_i N_{pool}$ samples from the N_i available, and likewise sampling $p_j N_{pool}$ samples (with replacement) from the N_j available, to obtain a test set with varietal proportion of nearly exactly (p_i, p_j) as prescribed. Then, truncating the model’s output from the original $K = 8$ classes to the 2 of interest, we measure the model’s predicted varietal proportion. Here we benefit from the prior knowledge about what varieties we expect, the predictions becomes slightly more difficult when mixing more varieties. We bootstrap this approach using $N_b = 400$ repetitions, to obtain N_b predictions for each true varietal proportion, allowing us to estimate uncertainty. We summarize results in Fig. 1, and report all 28 possible pairs in appendix, Fig. 9 and Fig. 10. We also summarize the 28 results in an average plot, to provide an intuition about the overall trend in Fig. 11

The downward trend can be understood as a statistical effect: for low proportions, the first variety (i) is under-represented, and the mistake of predicting i when the true variety is j is more frequent, because there are more grains of variety j . And vice-versa for large proportions.

The systematic affine trend of these plots makes it possible to calibrate these curves using labeled test data (coming from pure stand plots), to correct varietal proportion estimates in the ultimate unlabeled data, de-facto reducing the error to the shaded area shown in the plots. This shaded area itself comes from the finiteness of the test set.

4.5 Transfer Learning

Although labeled data is available by construction in our agricultural setup, it is interesting to study how much our trained networks can transfer efficiently to new years. For this study, we focus on our best performing network, ConvNeXt, and on the transfer from Y1→Y2. The most ambitious and challenging way to transfer is to directly apply a model trained on Y1 data, to the Y2 data, without any adaptation: this is what we may call direct transfer. When applying this direct transfer, we find our architectures to have about 33% balanced accuracy, a pretty catastrophic score. This negative result indicates that meteorological conditions have a large effect on the aspect of grains, such that the model cannot directly recognize the new year crop.

The classic way to perform transfer learning, still in a rather conservative way, is to use some Y2 labels to re-train the classification head of the neural network, keeping the backbone frozen. This requires labeled data of course, but only very light training. First, the frozen backbone is used to extract features (e.g. vectors of dimension 1280 for EfficientNet-V2) once and for all. One then trains

a simple logistic regression on top of it (which can be fitted exactly, being a single-layer network). This model has only $1280 \times 8 = 10240$ free parameters to be fitted on the Y2 data. We obtain a Bal-Acc of 89%, much closer to the upper bound given by the Y2 Bal-Acc of 94%. It’s worth noting here that to improve the training of the head, we use the Test Time Augmentations mentioned earlier: each training sampled is used 32 times (different views), and similarly for test samples. Without this trick, we get only about 75% accuracy.

All these performances are still statistically significantly lower than the Native Y2 performance, and this cannot be attributed to an intrinsic higher hardness of the Native Y2 task compared to the Native Y1 task. This result confirms the relevance of fine-tuning the whole network (backbone included), contrary to what is done in some studied where pre-trained models are only used as feature extractors.

5 Related Works

Grain variety classification is a canonical Fine-grained visual classification (FGVC) problem (Wah et al., 2011; Maji et al., 2013), as varieties share highly similar morphology, differing only in subtle textural cues, while the same variety may exhibit high intra-class variation across imaging conditions. Modern FGVC approaches combine multi-scale feature extraction, part-based attention, and contrastive pretraining to close the discriminability gap between visually similar classes. Recent applications include pest recognition across life stages (Han et al., 2024) or ship identification (Wang et al., 2024).

We are especially interested in the specific field of grain classification (sometimes referred to as kernel classification) using computer vision tools. There is a wealth of literature on the topic, and a lack of unified benchmark dataset to compare methods upon, which is however not the topic of the present paper.

Hyper-Spectral Imaging (HSI). A number of works (Liao et al., 2018; Qiu et al., 2018; Bao et al., 2019) make use of hyper-spectral input (HSI) to distinguish varieties of grains from the same species (maize, rice, wheat). Our goal here is to show that RGB inputs can be sufficient to classify with a high accuracy, yet we briefly comment on these works, as they share methodological points with our approach.

(Liao et al., 2018) use a pretrained VGGNet on HSI samples of maize seed to perform a relatively easy task, that of binary classification between haploid and diploid maize seeds, reaching up to 95% accuracy. (Qiu et al., 2018) study rice seed classification, focusing on the spectrum: for each grain, averaging all grain-belonging pixels into a meta-pixel, they apply 1D convolutional networks on the obtained spectrum, thus entirely discarding the spatial (textural) information, reaching up to 80% accuracy. In (Dreier et al., 2022), HSI images of 4 varieties of wheat grains densely packed are classified using a modified ResNet18 that can take multiple channels (more than 3) as input. Accuracies of up to 99% are reported. However, the models are not trained on the task of single-grain resolution, and their figure 10 actually reports considerable degradation of performance under a decrease of the density of grains in the input images. This can be understood as the patch-level error rate should decrease with grain number n as the single-grain error rate ε as $\approx \varepsilon^n$. So it is not clear whether that setup readily applies to our application of interest (where nearby grains may belong to different varieties).

RGB inputs, frozen backbone, extracted features. Some works, that we do not extensively discuss here, use a pre-trained frozen backbone to extract features from RGB inputs, applying some conventional ML methods on top to classify. For instance (Yasar and Golcuk, 2025) compares ResNet18, MobileNet-V2, DenseNet-201 and Inception-ResNet-V2 backbones and shows DenseNet+SVM to be optimal on their task. Our results, however, show that one can get systematically better results by fine-tuning the entire network (starting from pre-trained weights).

RGB inputs, training the backbone. (Javanmardi et al., 2021) classify corn seeds from 9

varieties with a standard CNN, reporting accuracy of up to 98%, without needing any particular architecture or training optimization. (Guo et al., 2024) design a specific CNN tailored to wheat grain classification, LWheatNet, that is shown to outperform classic architectures AlexNet, VGG16, MobileNet V2, MobileNet V3, and ShuffleNet V2. Pre-training is not mentioned so it is not clear whether authors leveraged this aspect of classic architectures. Also, worryingly, they mention that the dataset is build from taking two pictures of each grain at different angles, so the random train/test split makes it so some seeds appear both in train and test. (Mengstu and Taner, 2025) is probably the best work on the subject of RGB input grain classification that we know of. They propose a custom CNN to classify their newly introduced massive dataset of 124 bread wheat varieties comprising 465000 images (about 3750 images per class). Remarkably, they achieve 95% balanced accuracy on this 124-classes problem, their custom network outperforming classic architectures. Interestingly, they note how training time augmentations (multi-view) is crucial to good training, a fact our study confirms, but also oppose that characteristic of their pipeline with the use of pre-trained networks. Indeed, they study the cases of training from scratch (incurring 80 to 115h of GPU training time) or transfer-learning (re-training only the classification head) but do not attempt fine-tuning like we do (starting from pre-trained weights, yet tuning also backbone parameters).

Conclusion on related works: Although a large number of works study a problem similar to ours, it seems that none of them fully leverage the potential of pre-trained recent architectures. We further compare our work with previous ones in the conclusion.

6 Conclusion

In this paper we classify wheat grains from purely RGB input, using pre-trained models as the starting point of an efficient fine-tuning, making use of numerous training tricks, in particular aggressive train-time data augmentations. We reach 92% balanced accuracy after 5 minutes of fine-tuning and up to 94% with 10-15 minutes (training on GPU).

Our transfer-learning result confirms the relevance of fine-tuning the whole network (backbone included), contrary to what is done in some studies (Yasar and Golcuk, 2025) where pre-trained models are only used as feature extractors, or works where pre-trained models are seen as incompatible with train-time augmentations (Mengstu and Taner, 2025). Indeed, we find that fine-tuning systematically improves results over pure transfer, even when the fine-tuning is done on wheat grains of the very same varieties, only on a different year. This indicates that although robust, the models can always be improved by using labeled data produced under the exact same conditions as those targeted in the test set, even with relatively modest labeled dataset size (about 1600 grains per variety). The fine-tuning is made possible by our restriction from HSI to purely RGB input, as most vision foundation models take only 3 channels as input. Considering HSI input and HSI foundation models such as (Wang et al., 2025; Pang et al., 2024) or most notably (Braham et al., 2025), is left for future work.

Yet, these observations are also an encouragement to the training of a wheat grain classification foundation model. With a few more years of data and/or varieties of wheat, we could hope to make the backbone more generic and claim a foundation model. This is left for future work.

Last but not least, reviewing the literature, we noticed how in most papers (including this one, unfortunately), authors typically introduce their own grain dataset and compare a few models, but never try their new model on earlier datasets, making comparison of models impossible. Aggregating such datasets and systematically benchmarking various models and approaches on the same dataset(s) would greatly help to clarify the field. One could start with the datasets listed in (Mengstu and Taner, 2025). This is left for future work.

Our test data is, indirectly, available at Codabench: [will be made available upon publication] One can submit prediction codes and measure their performance on the held-out test set. A starting notebook is provided to facilitate onboarding and help participants get started quickly..

Our code is available at Zenodo: [will be made available upon publication].

333 Our training data is available at Zenodo: [will be made available upon publication]. The test set
334 will remain hidden, as long as the challenge is going on.

335 Authors Contributions

336 All authors have read and agreed to the published version of the manuscript.

337 Conceptualization: T.F. , F.P.L. , L.S. ; Literature Review: R.L. , F.P.L. ; Data Acquisition:
338 M.E. , T.F. ; Data Curation: M.E. , T.F. , F.P.L. , P.H.Q. ; Methodology: F.P.L. , L.S. , O.A.H. ,
339 O.O.P. , A.M. ; Software: O.A.H. , O.O.P. , A.M. ; Formal Analysis: O.A.H. ; Validation: O.A.H. ,
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A Methodological details

A.1 Dataset details

The 8 wheat varieties we study in this paper are: Accroc, Aubusson, Bagou, Belepi, Bergamo, Boregar, Expert, Kalahari, in this order. See Fig. 2, 3 and 4.

A.2 Training Details

Our training protocol is essentially the same for all the models we compare, up to slight variations in hyper-parameter values. Inputs are $224 \times 224 \times 3$, training runs for up to 50 epochs with early stopping on validation accuracy. As optimizer, we use AdamW with base learning rate of 3×10^{-4} , and weight decay $\beta = 10^{-2}$. The learning rate scheduler has a 5-epoch linear warm-up followed by cosine annealing down to 10^{-6} . On top of this base learning rate, we use layer-wise learning rate decay (LLRD): for each layer, weights are assigned a specific learning rate that increases with depth: early layers have a very small learning rate (remember that we are fine-tuning the network starting from pre-trained weights), up to the last layers that have a larger one. For ConvNeXt, we do not use LLRD.

To enhance model robustness and prevent overfitting we apply a comprehensive suite of data augmentations. Specifically each image undergoes random resized cropping (He et al., 2016) and D4 symmetry transformations (Vryniotis, 2021). Photometric distortions are introduced via color jittering which modifies brightness contrast saturation and hue (Vryniotis, 2021). We also introduce per-channel brightness jittering and additive Gaussian noise (Bishop, 1995) to simulate sensor variations. Furthermore, random erasing (Zhong et al., 2020) is applied to randomly mask rectangular regions of

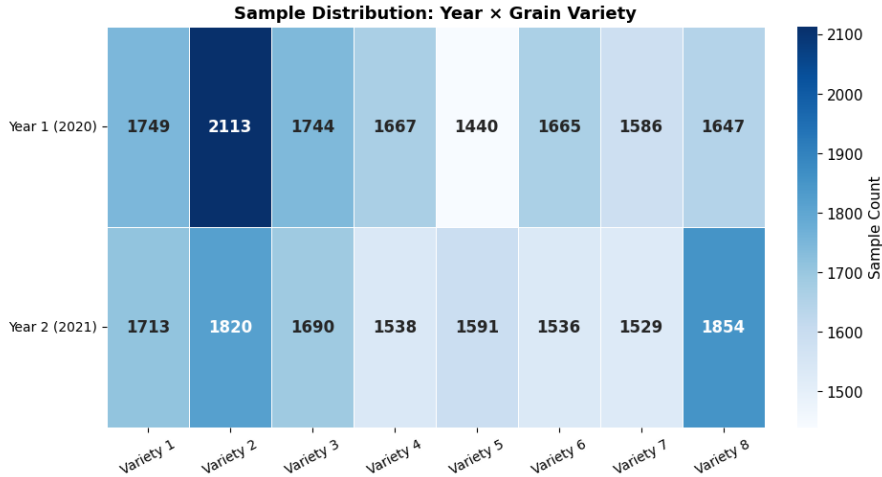


Figure 2: **Class distribution.** Sample counts per variety confirm approximately balanced classes in both years. Total per year is 13611 (Y1), 13271 (Y2), total: 26882.

In the paper we report balanced accuracies everywhere, but did not use weighted loss to cancel the moderate class imbalance we have.

the input and force the network to attend to the entire grain structure. As further augmentations, we use Mixup $\alpha = 0.4$ (Zhang et al., 2018) and CutMix $\alpha = 0.4$ (Yun et al., 2019), applied randomly with a 50 percent probability for each batch. We apply label smoothing with $\epsilon = 0.05$ (Szegedy et al., 2016) and utilize SWA (Izmailov et al., 2018), starting between epoch 5 and 8 depending on the specific model architecture to improve the model’s generalization. We use Automatic Mixed precision and gradient clipping ($\|\cdot\|_2 \leq 1.0$).

Thanks to these numerous augmentations, the effective training set size is boosted by several orders of magnitude, as the network receives as input various corrupted views of each training sample, in practice rarely seeing the exact same input twice over the ~ 20 epochs of training.

Each training of a model takes about 10 minutes on a NVIDIA RTX PRO 6000 Blackwell (Server Edition). Inference is much faster, and even on CPU, it takes between **0.80 ms** (ResNet-18) and **10.44 ms** (ViT-B/16) per sample, with our best models (EfficientNetV2-S, ConvNeXt-Tiny) in the **5.6–5.8 ms** range.

The train/test split is made with a ratio of 80-20 and using stratification by class, so as to keep the imbalance constant between sets. In practice, at train time, the training set is further split at random in train and validation sets.

When measuring test set accuracy, we use Test-Time Augmentations: for each test sample we perform a combination of 4 crops, 4 rotations (multiples of $\pi/4$) and 2 mirror symmetries, resulting in 32 views per test sample, whose softmax outputs are averaged before making a prediction. We do not use the most challenging (training time) augmentations, which makes prediction easier. This explains why the test accuracy is higher than the train accuracy and can be higher than the validation one.

B Results: details

B.1 Cross-Validation figures

B.2 Performance over training epochs and No Augmentation case

See Fig. 5.

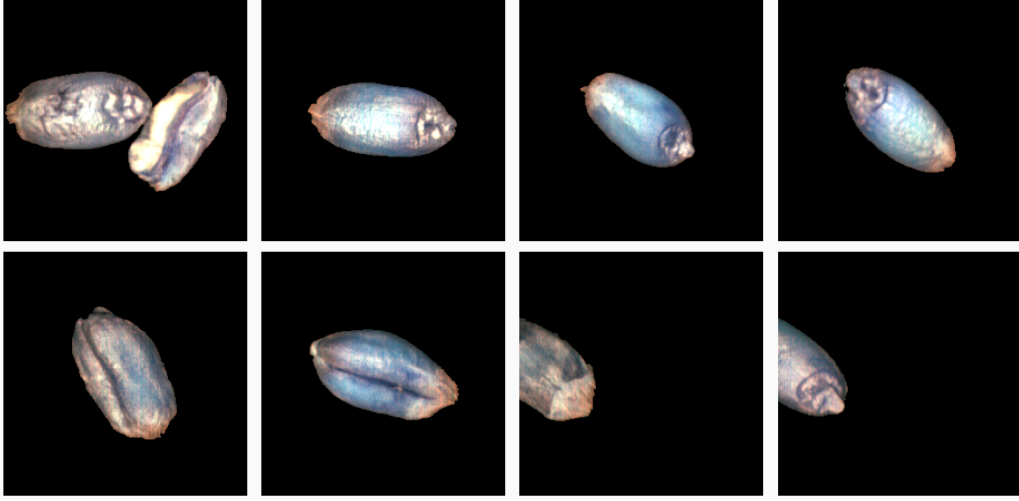


Figure 3: **Dataset excerpt (RGB images)**. Some grains are so close they are not separated by the watershed algorithm. Some grains may also lie at the boundary of the camera field of view. Grains can be one the furrow or back side, this does not matter. Here we cherry picked weird examples, but most grains are “normal”.

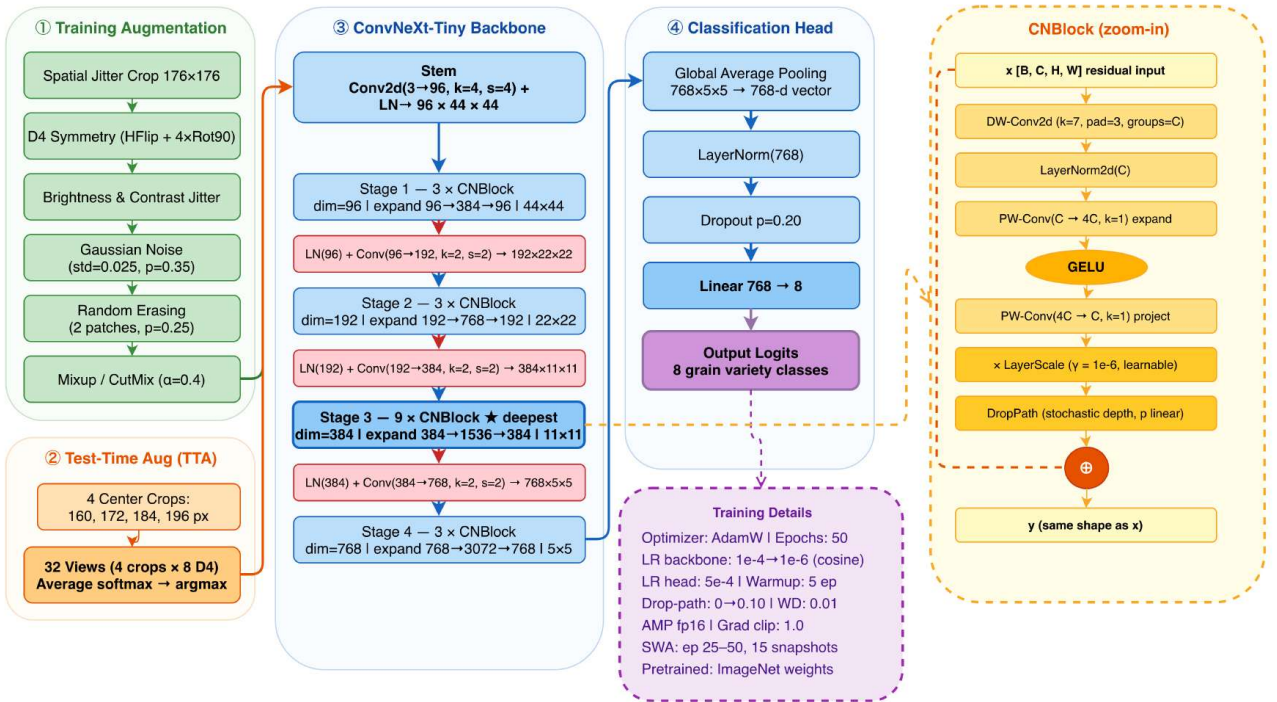


Figure 4: **Model Card and Training Pipeline.**

Model	Y1, 1	Y1, 2	Y1, 3	Y1, 4	Y1, 5	Y2, 1	Y2, 2	Y2, 3	Y2, 4	Y2, 5
ConvNeXt-Tiny	0.945	0.949	0.943	0.941	0.942	0.943	0.945	0.943	0.947	0.947
EfficientNetV2-S1	0.939	0.936	0.934	0.933	0.930	0.935	0.941	0.936	0.939	0.940
Swin-T1	0.933	0.936	0.937	0.933	0.936	0.93	0.926	0.936	0.930	0.933
ViT-B-16	0.903	0.910	0.911	0.905	0.911	0.910	0.91	0.909	0.911	0.914
ResNet-18	0.859	0.861	0.849	0.840	0.855	0.861	0.859	0.854	0.862	0.866

Table 2: **Cross-Validation detailed results.** We report the test set Bal-Acc for all models and both years, training (fine-tuning) models for only 50 epochs.

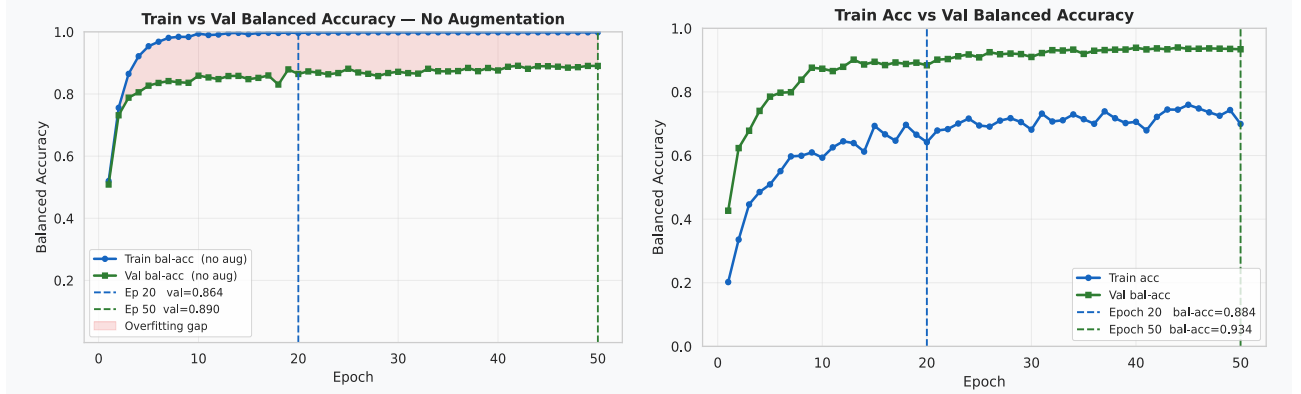


Figure 5: **No-Augmentation case & Performance dependence over time (epochs).** Left: training ConvNeXt on Native Y1 task with *all data augmentations turned off*. The test set Bal-Acc reaches about 90%, notably below the 94% reached using augmentations (with equal training time). We note that some overfitting is always present, and validation accuracy does not get as high as in the right panel.

Right: training ConvNeXt on Native Y1 task *with normal setup (aggressive data augmentations)*. Test Bal-Acc is at about 94%. Contrary to the usual case, training accuracy is worse than the validation one. In addition, validation accuracy is higher than on the left-side plot.

B.3 More ablation studies

B.4 Confusion matrices

See Fig. 6.

B.5 Calibration

See Fig. 7 and 8.

B.6 Varietal proportion estimates: all possible pairs

See Fig. 9, 10 and 11.

Experiment	Score	Δ
All ON	0.9384	0
No LR scheduler	0.9321	-0.0063
No Mixup/CutMix	0.9343	-0.0041
No label smoothing	0.9365	-0.0019
No Stochastic Weight Averaging (SWA)	0.9417	+0.0033
All OFF	0.9233	-0.0151

Table 3: **Training Tricks: ablation study.** Starting from the standard setup (top row), we turn off some other data augmentations and other learning tricks (SWA, TTA) one by one, and notice very little effect. When we turn off all of them (last row), the effects add up. Some noise can be due to the randomness of the fine-tuning process or the random seed of the dataloader, which can induce some fluctuations. Surprisingly, SWA seems to slightly deteriorate the test performance.

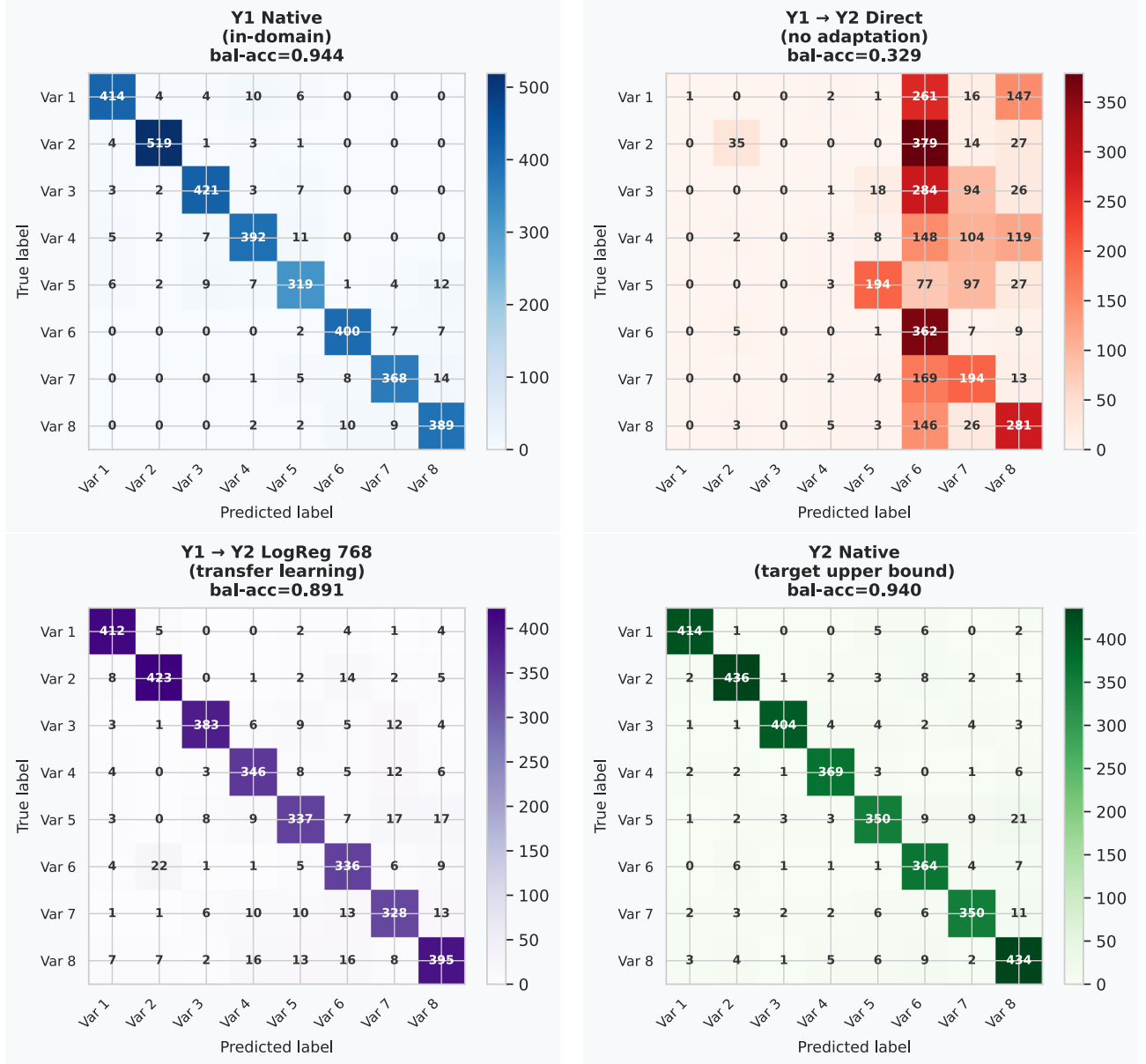


Figure 6: **Confusion matrices**, for ConvNeXt-Tiny for our 4 tasks: Native Y1, direct transfer Y1-Y2, transfer-learn Y1-Y2, and Native Y2. For direct transfer Y1-Y2, the model mostly predicts classes 6,7,8, resulting in an overall low balanced accuracy. For all other tasks, we note that the number of mistakes is low for all classes, and no particular confusions occur, showing that balanced accuracy is a valid choice of summary statistics.

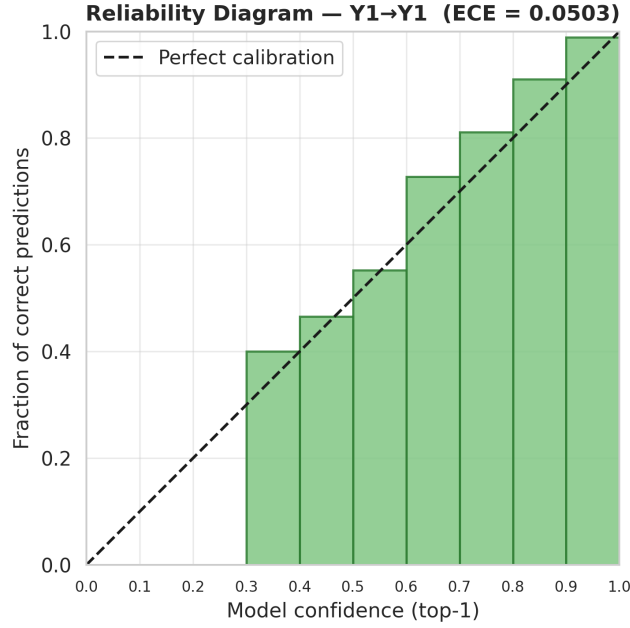


Figure 7: **Calibration quality.** ConvNeXt model on the Native Y1 task. For ranges of model confidence (largest logit) (x-axis) we record the fraction of correct predictions (8 classes classification). The alignment between confidence and empirical accuracy is excellent.

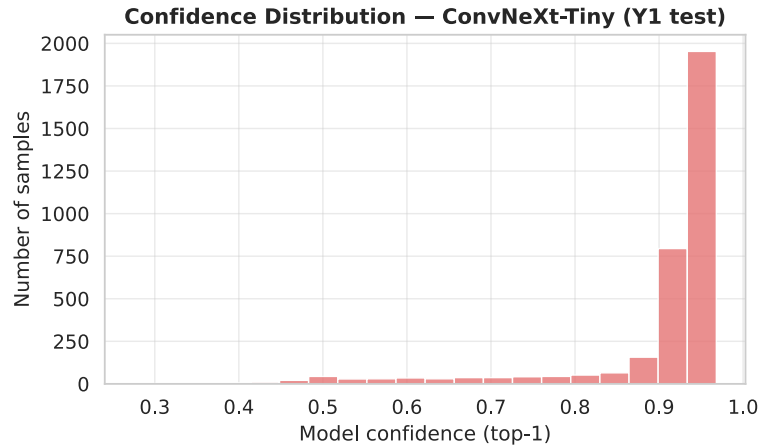


Figure 8: **Confidence distribution.** ConvNeXt model on the Native Y1 task. We report the distribution of logits, taking the largest logit for each sample. The confidence concentrates in the range $[0.9, 0.97]$.

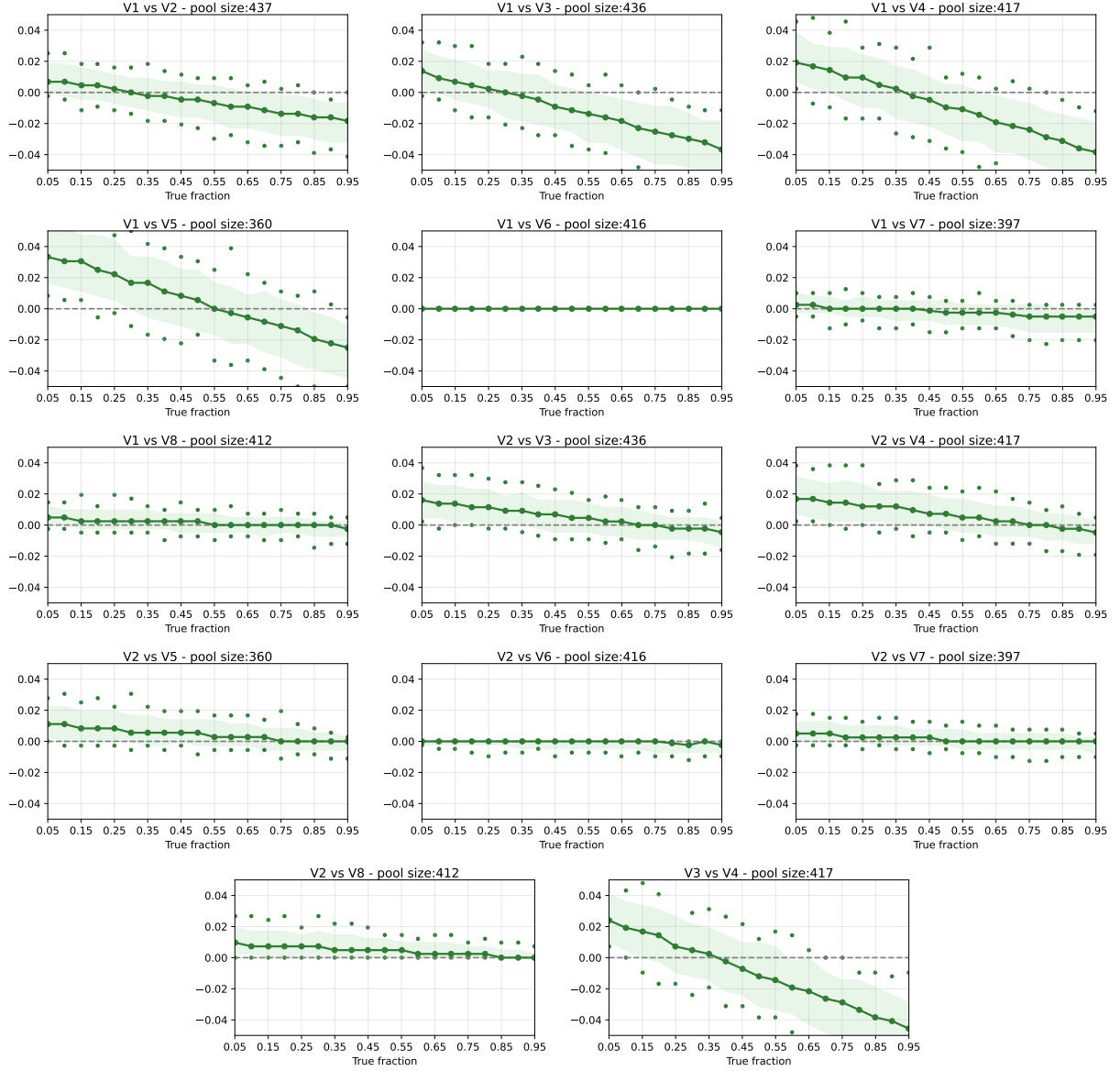


Figure 9: **Varietal proportion Estimates – all possible variety pairs, part 1.** A positive error indicates the first variety of the pair has been over-estimated. Bold line is the median over bootstrapped test sets. Shaded area indicates the 95% confidence interval. Min and Max value are shown as outlying smaller dots. Some pairs are perfectly estimated. The worse error is of about $\pm 4\%$ at the tips of the distribution, which is a large relative error in those cases, but the absolute error remains low.

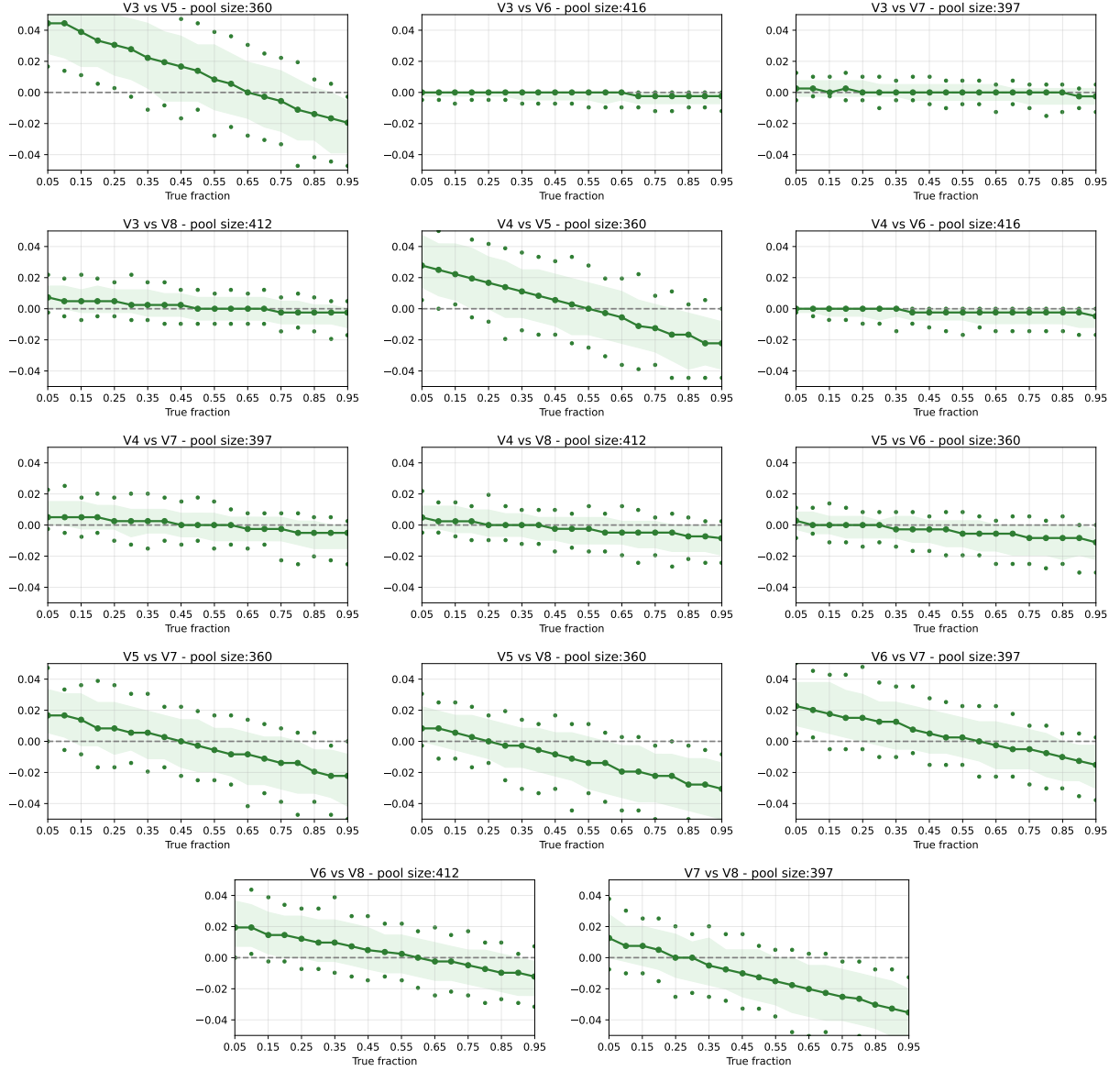


Figure 10: **Varietal proportion Estimates – all possible variety pairs, part 2.** A positive error indicates the first variety of the pair has been over-estimated. Bold line is the median over bootstrapped test sets. Shaded area indicates the 95% confidence interval. Min and Max value are shown as outlying smaller dots. Some pairs are perfectly estimated. The worse error is of about $\pm 4\%$ at the tips of the distribution, which is a large relative error in those cases, but the absolute error remains low.

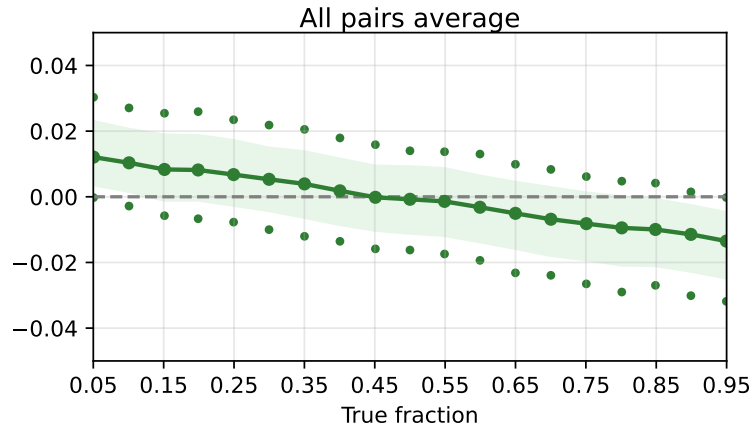


Figure 11: **Varietal proportion Estimates – average of all cases.** Here we make nonsensical averages, just to provide a summary of the trends seen in the 28 cases. We average the median error, confidence interval boundaries and min/max values.

Note that we also tried to weight predictions by their normalized logits, but this only deteriorates the results, because samples with low-confidence are not balanced among classes.